

Detailed Mathematical Formulation

Distributed Call-Center Staffing Optimization

Dual Service Metrics, QUBO Relaxation, and Hybrid Quantum–Classical Algorithms

Classical baselines · Pauli Correlation Encoding · CVaR-QAOA

Abstract

This document provides a complete mathematical formulation of the call-center staffing problem used in the hybrid quantum–classical notebooks. We define decision variables (unary binary encodings of shift headcounts), interval coverage, Erlang-C service metrics, hard and soft constraint systems, the official dual-feasible minimum-cost scoring rule, slack-free QUBO and Ising embeddings, Conditional Value-at-Risk Quantum Approximate Optimization Algorithm (CVaR-QAOA) with warm start, Pauli Correlation Encoding (PCE), classical repair operators, exact enumeration, and linear integer-programming surrogates. All symbols are defined; derivations of the quadratic objective and correlator capacity bounds are included.

Contents

1	Business problem and scoring rule	2
2	Sets, indices, and parameters	2
2.1	Index sets	2
2.2	Structural parameters	2
2.3	Demand and load	2
2.4	Penalty and dual parameters	2
3	Decision variables	3
3.1	Unary binary encoding	3
3.2	Coverage map	3
3.3	Derived totals	3
4	Erlang-C queueing model	3
4.1	Waiting probability	3
4.2	Service level	4
4.3	Average speed of answer	4
4.4	Occupancy and abandonment proxy	4
4.5	Requirement inversion	4
5	Hard constrained integer program	4
5.1	Primal formulation	4
5.2	Unary binary equivalent	5
5.3	True dual filter	5
5.4	Exact enumeration baseline	5
5.5	Linear ILP surrogate	5

6	Quadratic unconstrained relaxation (QUBO)	6
6.1	Shortfall-squared penalties	6
6.2	Expansion to QUBO matrix	6
6.3	Ising Hamiltonian	6
7	Warm-start CVaR-QAOA	6
7.1	Ansatz	6
7.2	CVaR objective	7
7.3	Dual averaging (optional outer loop)	7
7.4	Sampling and acceptance	7
8	Pauli Correlation Encoding (PCE)	7
8.1	Correlator capacity	7
8.2	Variable-to-Pauli map	8
8.3	Master loss	8
8.4	Compression ratio	8
9	Classical repair and local search	8
9.1	Coverage repair	8
9.2	ASA push	8
9.3	One-flip local search	8
9.4	Multi-start local search baseline	9
10	Greedy constructive baseline	9
11	Optimality gap and comparison metrics	9
12	Synthetic demand model	9
13	Manager control scenarios	9
14	Complexity and scalability remarks	10
15	Symbol glossary	10
16	Correspondence to notebooks	10

1. Business problem and scoring rule

A contact center must decide how many agents to place on each shift so that forecasted interval-level demand is served under contractual quality targets while controlling labor cost.

Definition 1 (Dual service targets). *Let $SL^* \in (0, 1)$ be the target service level (default 0.80) and $ASA^* > 0$ the target average speed of answer in seconds (default 25). A staffing plan is dual-feasible if its simulated service level satisfies $SLA \geq SL^*$ and its simulated average speed of answer satisfies $ASA \leq ASA^*$.*

Definition 2 (Official scoring rule). *Among all dual-feasible plans, the scored optimum is any plan of minimum total staffing and overtime cost. Ties may be broken by secondary criteria (e.g. utilization, abandonment), but cost is primary.*

This rule is the sole basis for ranking classical and quantum methods in the notebooks.

2. Sets, indices, and parameters

2.1. Index sets

$$\mathcal{S} = \{1, \dots, S\} \quad \text{shifts (e.g. Early, Mid, Late),} \quad (1)$$

$$\mathcal{T} = \{1, \dots, T\} \quad \text{time intervals (e.g. 30-minute buckets),} \quad (2)$$

$$\mathcal{K} = \{0, \dots, N_{\max} - 1\} \quad \text{unary slots per shift.} \quad (3)$$

2.2. Structural parameters

$$C_{s,t} \in \{0, 1\} \quad \text{coverage incidence: shift } s \text{ works interval } t, \quad (4)$$

$$c_s > 0 \quad \text{cost of one agent-shift (OT shifts scaled by factor } \rho_{\text{OT}}), \quad (5)$$

$$N_{\max} \in \mathbb{Z}_+ \quad \text{maximum agents allowed on any single shift,} \quad (6)$$

$$\Delta > 0 \quad \text{interval length in seconds,} \quad (7)$$

$$H > 0 \quad \text{average handle time (AHT) in seconds,} \quad (8)$$

$$\tau > 0 \quad \text{SLA answer-time threshold in seconds.} \quad (9)$$

Breaks are encoded by setting $C_{s,t} = 0$ on the break interval of shift s . Overtime shifts use $c_s \leftarrow \rho_{\text{OT}} c_s$ with typical $\rho_{\text{OT}} = 1.5$.

2.3. Demand and load

Let $\lambda_t \geq 0$ be the forecasted number of arrivals in interval t . The offered load (Erlangs) is

$$A_t = \frac{\lambda_t}{\Delta} H. \quad (10)$$

When arrivals are Poisson with rate $\mu_t = \lambda_t/\Delta$, A_t is the mean number of busy servers in an infinite-server system with mean service time H .

2.4. Penalty and dual parameters

$$P_{\text{SLA}}, P_{\text{ASA}} > 0 \quad \text{QUBO shortfall penalty weights,} \quad (11)$$

$$\lambda_t^{\text{dual}} \geq 0 \quad \text{optional Lagrange multipliers on SLA shortfalls,} \quad (12)$$

$$w_t \geq 0 \quad \text{optional ASA shortfall weights (may depend on } R_t^{\text{ASA}}). \quad (13)$$

3. Decision variables

3.1. Unary binary encoding

For each shift $s \in \mathcal{S}$ and slot $k \in \mathcal{K}$,

$$x_{s,k} \in \{0, 1\}. \quad (14)$$

The integer headcount on shift s is the sum of unary bits:

$$n_s = \sum_{k \in \mathcal{K}} x_{s,k} \in \{0, 1, \dots, N_{\max}\}. \quad (15)$$

The total number of binary decision variables is

$$N = |\mathcal{S}| \cdot N_{\max} = S \cdot N_{\max}. \quad (16)$$

In the standard small instance, $S = 3$, $N_{\max} = 4$, hence $N = 12$.

Remark 1 (Why unary?). *Unary encoding makes headcount penalties linear (or quadratic after coverage expansion) without requiring integer variables or log-encoding slack bits. It is convenient for QUBO/Ising embeddings and for warm-start from classical integer solutions.*

3.2. Coverage map

Staffing available in interval t is the linear combination

$$\text{cov}_t = \sum_{s \in \mathcal{S}} C_{s,t} n_s = \sum_{s \in \mathcal{S}} \sum_{k \in \mathcal{K}} C_{s,t} x_{s,k}. \quad (17)$$

In matrix form, if $\mathbf{n} = (n_1, \dots, n_S)^\top$ and $C = (C_{s,t}) \in \{0, 1\}^{S \times T}$,

$$\mathbf{cov} = C^\top \mathbf{n}. \quad (18)$$

3.3. Derived totals

$$\text{Total agents} = \sum_s n_s, \quad (19)$$

$$\text{Total cost} = \sum_s c_s n_s. \quad (20)$$

4. Erlang-C queueing model

4.1. Waiting probability

Consider n identical agents and offered load A . The steady-state probability that an arriving call must wait (Erlang-C formula) is

$$C(n, A) = \begin{cases} \frac{\frac{A^n}{n!} \frac{n}{n-A}}{\sum_{k=0}^{n-1} \frac{A^k}{k!} + \frac{A^n}{n!} \frac{n}{n-A}} & \text{if } n > A, \\ 1 & \text{if } n \leq A. \end{cases} \quad (21)$$

An equivalent recursive form used in implementation is

$$B_0 = 1, \quad B_k = 1 + \frac{k}{A} B_{k-1} \quad (k = 1, \dots, n), \quad C(n, A) = \frac{1}{1 + (1 - \rho)(B_n - 1)/\rho}, \quad (22)$$

where $\rho = A/n$ for $n > A$.

4.2. Service level

The probability that wait time does not exceed threshold τ is

$$\text{SL}(n, \lambda) = 1 - C(n, A) \exp\left(-(n - A) \frac{\tau}{H}\right), \quad (23)$$

with A as in (10). If $n \leq A$ or $n = 0$, we set $\text{SL} = 0$ (except the empty-demand case $\lambda = 0$, where $\text{SL} = 1$).

4.3. Average speed of answer

$$\text{ASA}(n, \lambda) = \frac{C(n, A) H}{n - A} \quad (n > A), \quad (24)$$

and $\text{ASA} = \infty$ (numerically a large sentinel) when $n \leq A$.

4.4. Occupancy and abandonment proxy

$$\text{occupancy}(n, \lambda) = \min(1, A/n), \quad (25)$$

$$\text{abandon}(n, \lambda) = \min(0.5, \alpha_0 + \alpha_1 \max(0, \text{ASA} - \beta)), \quad (26)$$

with constants $(\alpha_0, \alpha_1, \beta)$ fixed in the notebooks (e.g. base abandonment 0.02).

4.5. Requirement inversion

For each interval t , the minimum integer staffing meeting each target is

$$R_t^{\text{SLA}} = \min\{n \in \mathbb{Z}_+ : \text{SL}(n, \lambda_t) \geq \text{SL}^*\}, \quad (27)$$

$$R_t^{\text{ASA}} = \min\{n \in \mathbb{Z}_+ : \text{ASA}(n, \lambda_t) \leq \text{ASA}^*\}. \quad (28)$$

Both are computed by sequential search from a lower bound $\lceil A_t \rceil$.

Proposition 1 (Monotonicity). *For fixed λ , $\text{SL}(n, \lambda)$ is nondecreasing in n and $\text{ASA}(n, \lambda)$ is nonincreasing in n on $\{n : n > A\}$. Consequently R_t^{SLA} and R_t^{ASA} are well-defined and finite whenever a finite n meets the targets.*

5. Hard constrained integer program

5.1. Primal formulation

$$\min_{\mathbf{n}} \sum_{s \in \mathcal{S}} c_s n_s \quad (29)$$

$$\text{subject to } \sum_s C_{s,t} n_s \geq R_t^{\text{SLA}}, \quad \forall t \in \mathcal{T}, \quad (30)$$

$$\sum_s C_{s,t} n_s \geq R_t^{\text{ASA}}, \quad \forall t \in \mathcal{T}, \quad (31)$$

$$0 \leq n_s \leq N_{\max}, \quad n_s \in \mathbb{Z}, \quad \forall s \in \mathcal{S}. \quad (32)$$

5.2. Unary binary equivalent

Substituting (15),

$$\min_{\mathbf{x}} \quad \sum_s \sum_k c_s x_{s,k} \quad (33)$$

$$\text{s.t.} \quad \sum_s \sum_k C_{s,t} x_{s,k} \geq R_t^{\text{SLA}}, \quad \forall t, \quad (34)$$

$$\sum_s \sum_k C_{s,t} x_{s,k} \geq R_t^{\text{ASA}}, \quad \forall t, \quad (35)$$

$$x_{s,k} \in \{0, 1\}. \quad (36)$$

5.3. True dual filter

After any candidate $\mathbf{n}$ is produced, define the weighted (call-volume) averages

$$\text{SLA}(\mathbf{n}) = \frac{\sum_t \lambda_t \text{SL}(\text{cov}_t, \lambda_t)}{\sum_t \lambda_t}, \quad (37)$$

$$\text{ASA}(\mathbf{n}) = \frac{\sum_t \lambda_t \text{ASA}(\text{cov}_t, \lambda_t)}{\sum_t \lambda_t}. \quad (38)$$

The plan is dual-feasible iff

$$\text{SLA}(\mathbf{n}) \geq \text{SL}^* \quad \text{and} \quad \text{ASA}(\mathbf{n}) \leq \text{ASA}^*. \quad (39)$$

Constraints (30)–(31) are *sufficient but not necessary* for (39); the true filter uses the nonlinear maps.

5.4. Exact enumeration baseline

The integer box $\{0, \dots, N_{\max}\}^S$ has cardinality

$$|\mathcal{N}| = (N_{\max} + 1)^S. \quad (40)$$

For $S = 3$, $N_{\max} = 4$, $|\mathcal{N}| = 125$. Let

$$\mathcal{F} = \{\mathbf{n} \in \mathcal{N} : \mathbf{n} \text{ satisfies (39)}\}. \quad (41)$$

The exact scored optimum is

$$\mathbf{n}^* \in \arg \min_{\mathbf{n} \in \mathcal{F}} \sum_s c_s n_s, \quad (42)$$

obtained by exhaustive evaluation of SLA, ASA, and cost on every vector.

5.5. Linear ILP surrogate

Replace (30)–(31) by the single linear cover

$$\sum_s C_{s,t} n_s \geq \max(R_t^{\text{SLA}}, R_t^{\text{ASA}}), \quad \forall t, \quad (43)$$

and solve the resulting binary ILP with a MIP solver (e.g. CBC). Re-score with (37)–(38). Because linear R -covers ignore Erlang curvature, the ILP optimum can be strictly more expensive than $\mathbf{n}^*$.

6. Quadratic unconstrained relaxation (QUBO)

6.1. Shortfall-squared penalties

Define the soft objective

$$\begin{aligned} h(\mathbf{x}) = & \sum_s \sum_k c_s x_{s,k} \\ & + P_{\text{SLA}} \sum_t \lambda_t^{\text{dual}} \left(\text{cov}_t - R_t^{\text{SLA}} \right)^2 \\ & + P_{\text{ASA}} \sum_t w_t \left(\text{cov}_t - R_t^{\text{ASA}} \right)^2. \end{aligned} \quad (44)$$

No explicit feasibility indicator is required: when $\text{cov}_t \geq R_t$, the corresponding squared term need not be zero if one expands $(\text{cov} - R)^2$ fully, but the linear cost term discourages over-staffing. In practice the notebooks expand

$$(\text{cov}_t - R_t)^2 = \text{cov}_t^2 - 2R_t \text{cov}_t + R_t^2 \quad (45)$$

and absorb the linear piece into coefficients of $x_{s,k}$, using $x_{s,k}^2 = x_{s,k}$.

6.2. Expansion to QUBO matrix

Write $\text{cov}_t = \sum_{i \in \mathcal{I}_t} x_i$ where $\mathcal{I}_t$ indexes unary bits covering interval t . Then

$$\text{cov}_t^2 = \sum_{i \in \mathcal{I}_t} x_i + 2 \sum_{\substack{i < j \\ i, j \in \mathcal{I}_t}} x_i x_j. \quad (46)$$

Collecting terms yields

$$h(\mathbf{x}) = \sum_i Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j + \text{const}, \quad (47)$$

or in matrix form $\mathbf{x}^\top Q \mathbf{x}$ with the usual identification of the linear part on the diagonal for binary $\mathbf{x}$.

Proposition 2 (Slack-free encoding). *The formulation (44)–(47) introduces no auxiliary slack variables. The number of binary variables remains $N = S \cdot N_{\text{max}}$.*

6.3. Ising Hamiltonian

Map $x_i = (1 - Z_i)/2$ with Pauli Z_i . Substituting into (47) gives

$$H = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j + E_0, \quad (48)$$

where the coefficients (h_i, J_{ij}, E_0) are affine functions of the QUBO entries. Ground states of H correspond to minimizers of h .

7. Warm-start CVaR-QAOA

7.1. Ansatz

Let $p \geq 1$ be the number of layers. The warm-started QAOA state is

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \left(\prod_{\ell=1}^p e^{-i\beta_\ell B} e^{-i\gamma_\ell H} \right) |\psi_0\rangle, \quad (49)$$

where $B = \sum_i X_i$ (or a warm-start mixer that conjugates a biased initial state) and $|\psi_0\rangle$ is prepared by single-qubit rotations

$$|\psi_0\rangle = \bigotimes_{i=1}^N R_Y(\theta_i)|0\rangle, \quad \theta_i = 2 \arcsin \sqrt{\tilde{x}_i}, \quad (50)$$

with $\tilde{x}_i \in (\varepsilon, 1 - \varepsilon)$ a softened classical bit (e.g. from greedy).

7.2. CVaR objective

Let E be the random energy obtained by measuring $|\psi\rangle$ in the computational basis and evaluating H (or h). For risk level $\alpha \in (0, 1]$,

$$\text{CVaR}_\alpha(E) = \frac{1}{\alpha} \int_0^\alpha F_E^{-1}(u) \, du = \frac{1}{\alpha} \sum_{j: F(E_{(j)}) \leq \alpha} p_{(j)} E_{(j)}, \quad (51)$$

where $E_{(j)}$ are ordered sample energies. The classical outer loop minimizes CVaR_α over (γ, β) (e.g. COBYLA).

7.3. Dual averaging (optional outer loop)

Given residual shortfalls $r_t = \max(0, R_t^{\text{SLA}} - \text{cov}_t)$ from the best sample of outer iteration k ,

$$\lambda_t^{\text{dual}, (k+1)} = \max\left(0, \lambda_t^{\text{dual}, (0)} + \eta \cdot \frac{1}{k} \sum_{j=1}^k r_t^{(j)}\right). \quad (52)$$

Updated multipliers re-weight the QUBO (44) for the next quantum solve.

7.4. Sampling and acceptance

Draw M bitstrings; for each, apply classical repair (Section 9); retain the dual-feasible plan of minimum cost. If none is dual-feasible, retain the best SLA-feasible plan or report failure.

8. Pauli Correlation Encoding (PCE)

8.1. Correlator capacity

Definition 3 (Degree- D Z-correlators). *On n_q qubits, the set of Pauli strings of weight at most D using only Z is*

$$\mathcal{P}_D(n_q) = \{Z_{i_1} \cdots Z_{i_d} : 1 \leq d \leq D, 1 \leq i_1 < \cdots < i_d \leq n_q\}. \quad (53)$$

Its cardinality is

$$|\mathcal{P}_D(n_q)| = \sum_{d=1}^D \binom{n_q}{d}. \quad (54)$$

Proposition 3 (Minimum compressed width). *The smallest register size supporting an injective map from N variables to degree- D correlators is*

$$n_q^*(N, D) = \min\left\{q \in \mathbb{Z}_+ : \sum_{d=1}^D \binom{q}{d} \geq N\right\}. \quad (55)$$

For $D = 2$, $N = 12$, one obtains $n_q^* = 5$ because $\binom{5}{1} + \binom{5}{2} = 15 \geq 12$.

8.2. Variable-to-Pauli map

Fix an injection $\pi : \{1, \dots, N\} \rightarrow \mathcal{P}_D(n_q)$. Write $P_i = \pi(i)$. Soft bits are

$$x_i^{\text{soft}}(\theta) = \frac{1}{2} \left(1 - \tanh(\alpha \langle \psi(\theta) | P_i | \psi(\theta) \rangle) \right), \quad (56)$$

with $\alpha > 0$ a decode sharpness and $|\psi(\theta)\rangle$ an EfficientSU2 (or similar) ansatz on n_q qubits.

8.3. Master loss

$$L(\theta) = h(\mathbf{x}^{\text{soft}}(\theta)), \quad (57)$$

optimized by a classical derivative-free method (COBYLA). Hard decode uses

$$\hat{x}_i = \begin{cases} 0 & \text{if } \langle P_i \rangle \geq 0, \\ 1 & \text{if } \langle P_i \rangle < 0. \end{cases} \quad (58)$$

8.4. Compression ratio

$$\text{qubit reduction} = 1 - \frac{n_q^*}{N}. \quad (59)$$

On the standard instance this is $1 - 5/12 \approx 58\%$.

9. Classical repair and local search

9.1. Coverage repair

Given $\mathbf{n}$ (or bits aggregated to $\mathbf{n}$):

$$\begin{aligned} &\textbf{while } \exists t : \text{cov}_t < R_t^{\text{SLA}} \textbf{ do} \\ &\quad t^* \leftarrow \arg \max_t (R_t^{\text{SLA}} - \text{cov}_t) \\ &\quad s^* \leftarrow \text{any } s \text{ with } n_s < N_{\max}, C_{s,t^*} = 1 \\ &\quad n_{s^*} \leftarrow n_{s^*} + 1. \end{aligned} \quad (60)$$

9.2. ASA push

$$\begin{aligned} &\textbf{while } \max_t \text{ASA}(\text{cov}_t, \lambda_t) > \text{ASA}^* \textbf{ do} \\ &\quad t^* \leftarrow \arg \max_t \text{ASA}(\text{cov}_t, \lambda_t) \\ &\quad s^* \leftarrow \text{feasible covering shift with spare capacity} \\ &\quad n_{s^*} \leftarrow n_{s^*} + 1. \end{aligned} \quad (61)$$

9.3. One-flip local search

From a dual-feasible $\mathbf{x}$, try bit flips:

$$\mathbf{x}' = \mathbf{x} + \mathbf{e}_i - 2x_i \mathbf{e}_i \quad (i = 1, \dots, N). \quad (62)$$

Accept the first (or best) neighbor that remains dual-feasible and strictly reduces cost. Iterate until local optimality.

9.4. Multi-start local search baseline

Generate starts from greedy and from uniform random $\mathbf{n} \in \{0, \dots, N_{\max}\}^S$; apply repair and one-flip (or agent-drop) search; keep the global best dual-feasible cost. On the 125-vector instance this recovers $\mathbf{n}^*$ in the notebooks.

10. Greedy constructive baseline

Initialize $\mathbf{n} = \mathbf{0}$ and residual requirement $\mathbf{r} = \max(\mathbf{R}^{\text{SLA}}, \mathbf{R}^{\text{ASA}})$. While $\|\mathbf{r}\|_1 > 0$ and capacity remains,

$$s^* = \arg \max_{s: n_s < N_{\max}} \frac{\sum_t C_{s,t} [r_t > 0]}{c_s}, \quad n_{s^*} \leftarrow n_{s^*} + 1, \quad \mathbf{r} \leftarrow \max(\mathbf{0}, \mathbf{r} - C_{s^*,:}). \quad (63)$$

Then apply (60)–(61). Greedy is dual-feasible but typically not optimal (about 12% cost gap to $\mathbf{n}^*$ on the standard instance).

11. Optimality gap and comparison metrics

For a method M returning dual-feasible $\mathbf{n}^M$,

$$\text{gap}(M) = \sum_s c_s n_s^M - \sum_s c_s n_s^*, \quad \text{gap}_{\%}(M) = 100 \cdot \frac{\text{gap}(M)}{\sum_s c_s n_s^*}. \quad (64)$$

Additional reported metrics:

$$\text{SLA}^M = \text{SLA}(\mathbf{n}^M), \quad \text{ASA}^M = \text{ASA}(\mathbf{n}^M), \quad (65)$$

$$\text{util}^M = \frac{1}{T} \sum_t \text{occupancy}(\text{cov}_t^M, \lambda_t), \quad \text{abandon}^M = \text{weighted mean abandonment}. \quad (66)$$

Runtime, qubit count, circuit depth, and shot budget are reported for quantum methods but do not enter the official cost ranking.

12. Synthetic demand model

Interval arrivals are generated as

$$\lambda_{d,t,q,c} \sim \text{Poisson}(\Lambda \cdot u_t \cdot \delta_d \cdot \gamma_q \cdot \kappa_c), \quad (67)$$

where u_t is a bimodal time-of-day profile, δ_d a day-of-week scale, γ_q a queue factor, and κ_c a channel factor (voice vs. chat). The forecast used by the optimizer is the multi-day mean

$$\hat{\lambda}_t = \frac{1}{D} \sum_{d=1}^D \sum_{q,c} \lambda_{d,t,q,c}. \quad (68)$$

Requirement tables $R_t^{\text{SLA}}, R_t^{\text{ASA}}$ are built from $\hat{\lambda}_t$ via (27)–(28).

13. Manager control scenarios

Scenario policies reweight the same model:

$$\text{Cost mode: } P_{\text{SLA}}, P_{\text{ASA}} \text{ reduced}, \quad (69)$$

$$\text{Service mode: } P_{\text{SLA}}, P_{\text{ASA}} \text{ increased}, \quad (70)$$

$$\text{Preference mode: } c_s \text{ tilted toward preferred shifts}, \quad (71)$$

$$\text{Resilience mode: } R_t \leftarrow R_t + 1 \text{ (buffer agents)}. \quad (72)$$

Each scenario is solved with the same hybrid or classical pipeline; dual feasibility is always enforced at acceptance.

14. Complexity and scalability remarks

- Exact enumeration is $O((N_{\max} + 1)^S \cdot T)$ Erlang evaluations — practical only for small S .
- Linear ILP has N binaries and $O(T)$ cover inequalities; solvable by branch-and-cut for moderate N .
- Direct QAOA uses N qubits; PCE reduces to $n_q^*(N, 2)$ qubits at the cost of correlator estimation.
- Enterprise scale (many skills, 48 intervals, large $N_{\max}$) requires decomposition (time blocks, skill groups, column generation) around the same QUBO core.

15. Symbol glossary

Symbol	Meaning
$x_{s,k}$	Unary binary: k -th agent slot on shift s
n_s	Integer agents on shift s
cov_t	Agents covering interval t
$C_{s,t}$	Shift–interval incidence (break $\Rightarrow 0$)
c_s	Cost per agent on shift s
λ_t, A_t	Arrivals and offered load in interval t
$C(n, A)$	Erlang-C waiting probability
SL, ASA	Service level and average speed of answer
$R_t^{\text{SLA}}, R_t^{\text{ASA}}$	Min. agents meeting each target
$h(\mathbf{x}), H$	QUBO objective and Ising Hamiltonian
CVaR_α	Conditional value-at-risk of energy
$P_i, \langle P_i \rangle$	PCE Pauli correlator and expectation
$\mathbf{n}^*, \text{gap}$	Exact dual-feasible optimum and cost gap

16. Correspondence to notebooks

Notebook	Equations primarily used
Classical baselines	(29)–(42), greedy (63), repair (60)–(61)
PCE staffing	QUBO (44), capacity (54)–(55), soft/hard (56)–(58)
Hybrid CVaR-QAOA	Ising (48), ansatz (49), CVaR (51), dual filter (39)

All solvers are evaluated under the same dual-feasible minimum-cost rule. Quantum procedures are hybrid: variational sampling followed by classical feasibility repair and optional local search.